Cornerstone of a building

Cornerstone in dental morphology?

PERMANENT MAXILLARY AND MANDIBULAR CANINES

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Lecturer | Oral Biology

LEARNING OBJECTIVES

To identify the positive and negative anatomical landmarks of permanent canines.

To distinguish maxillary canine from a mandibular canine.

To be able to draw the graph outline.

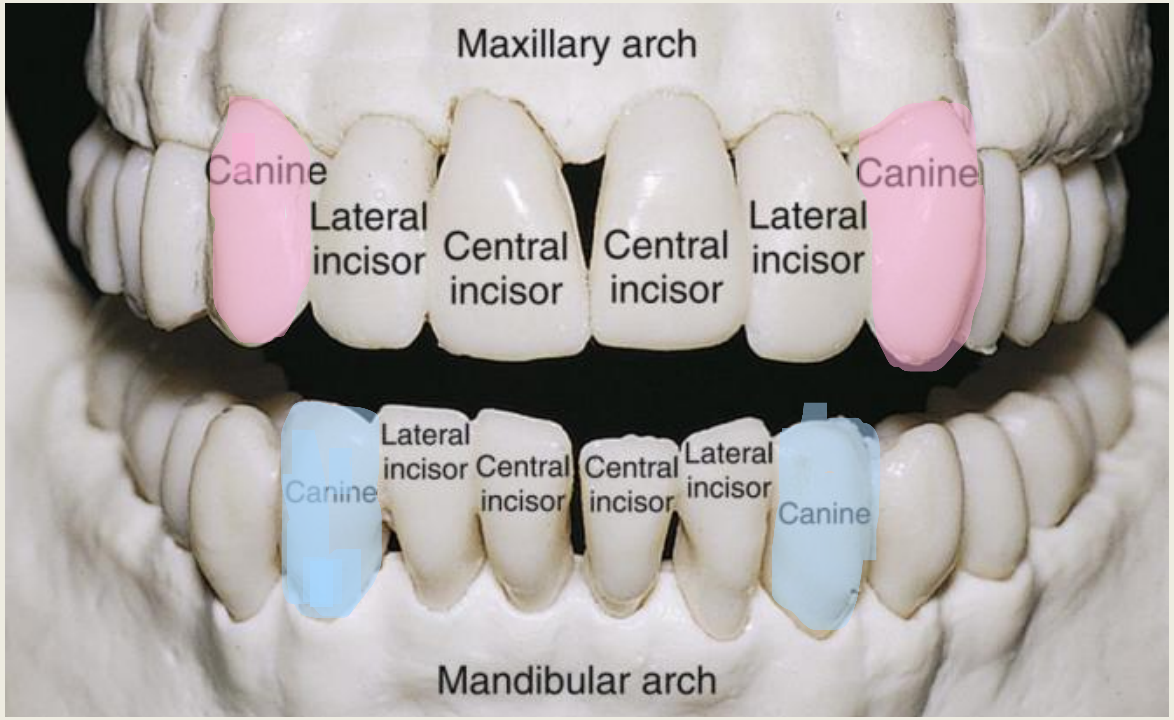
To be able to carve the canine tooth on wax models.

OVERVIEW OF CANINES

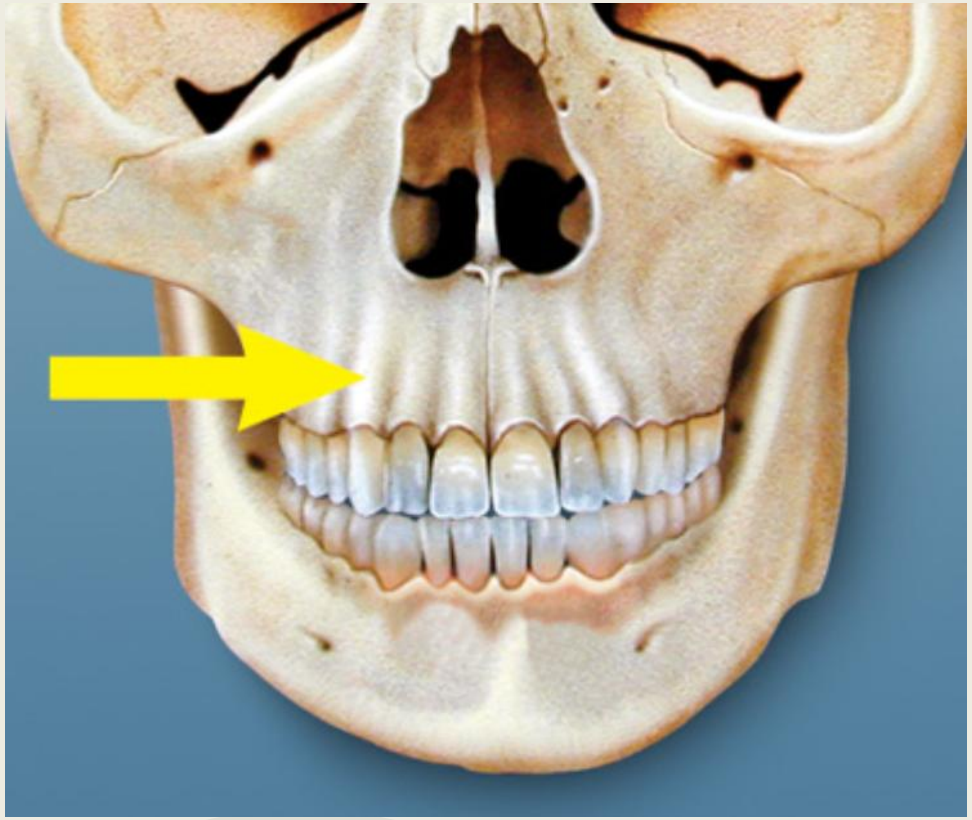
The four canines are placed at the “corners” of the mouth—**cornerstone** of the dental arches.

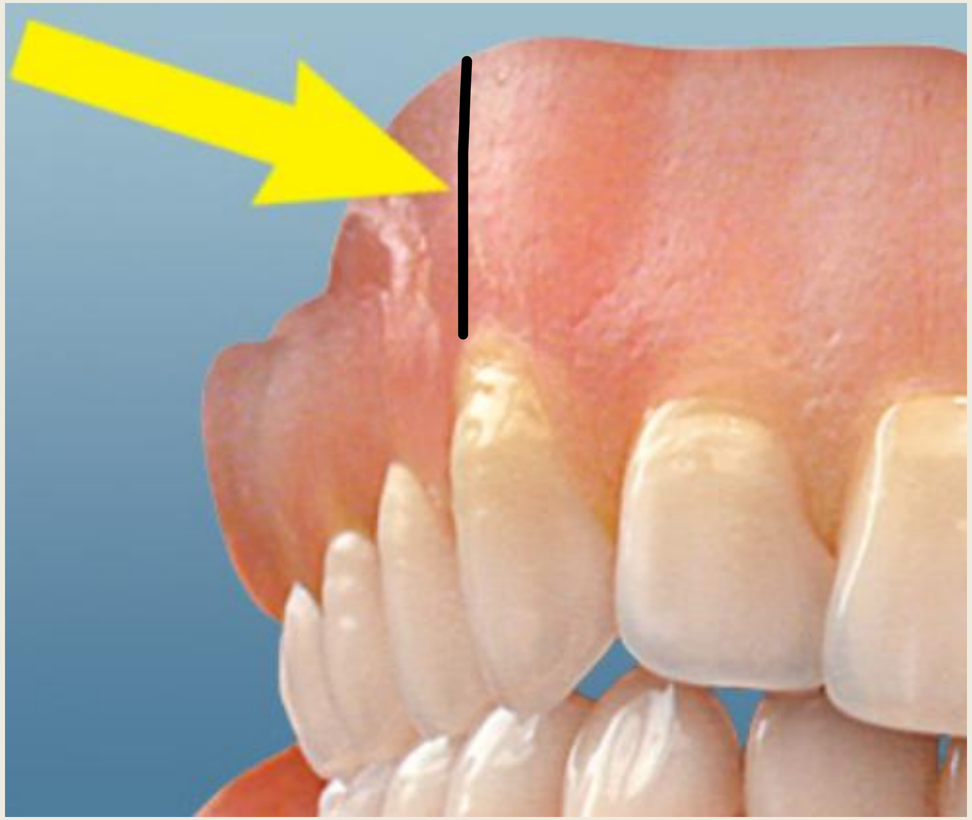
Dimensionally it is the longest tooth.

In functional form, bears a resemblance to incisor form and the premolar form.



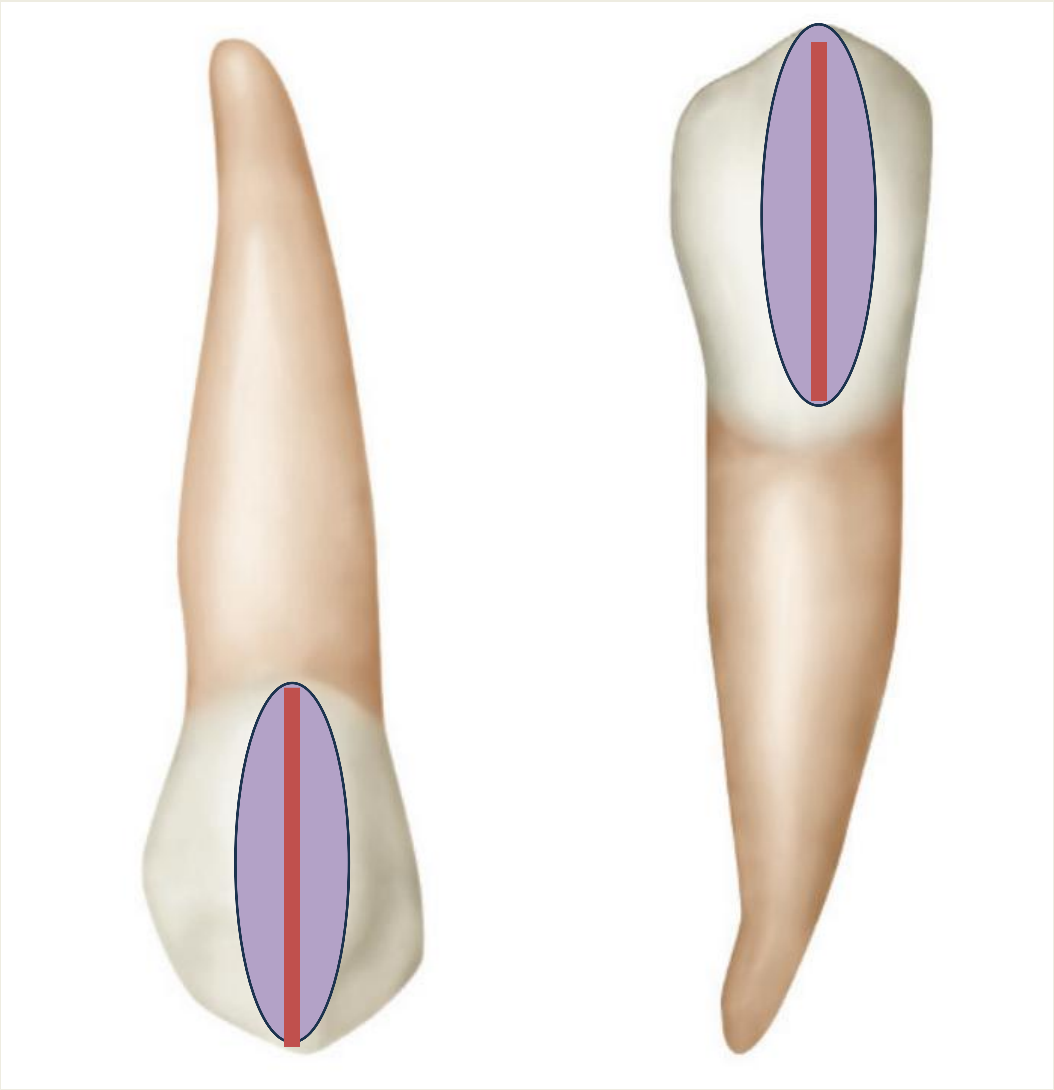
The positions of canines and their anchorage in the bone, along with the bone ridge over the labial portions of the roots, called the **canine eminence**, have a cosmetic value (help in normal facial expressions)





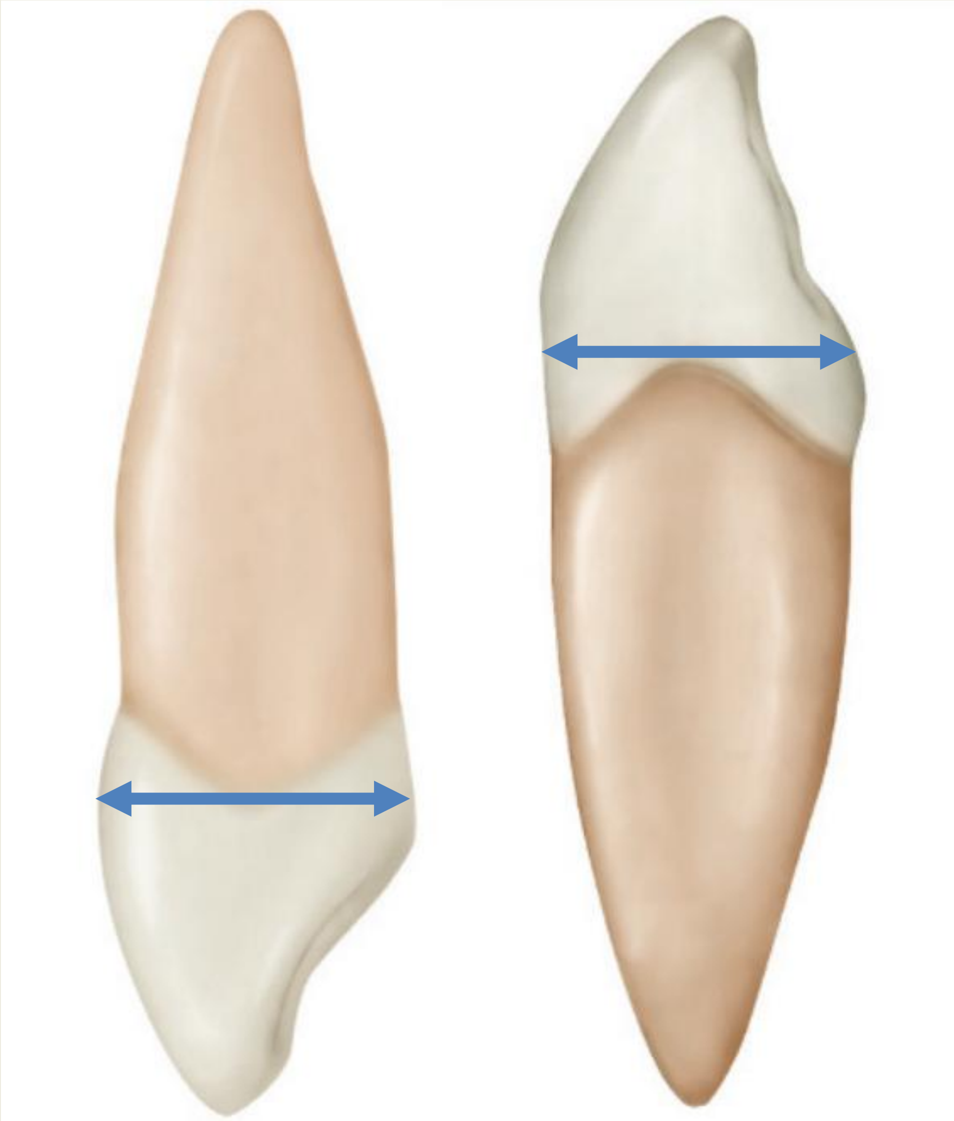
The middle labial lobes are highly developed into well-formed cusp—**cuspids**.

Canines are the only teeth with a **labial ridge.**



MAXILLARY **MANDIBULAR**

The measurement of canine crown is greater labiolingually than it is mesiodistally



MAXILLARY

MANDIBULAR

MORPHOLOGY OF MAXILLARY CANINES

CHRONOLOGY OF MAXILLARY CANINE

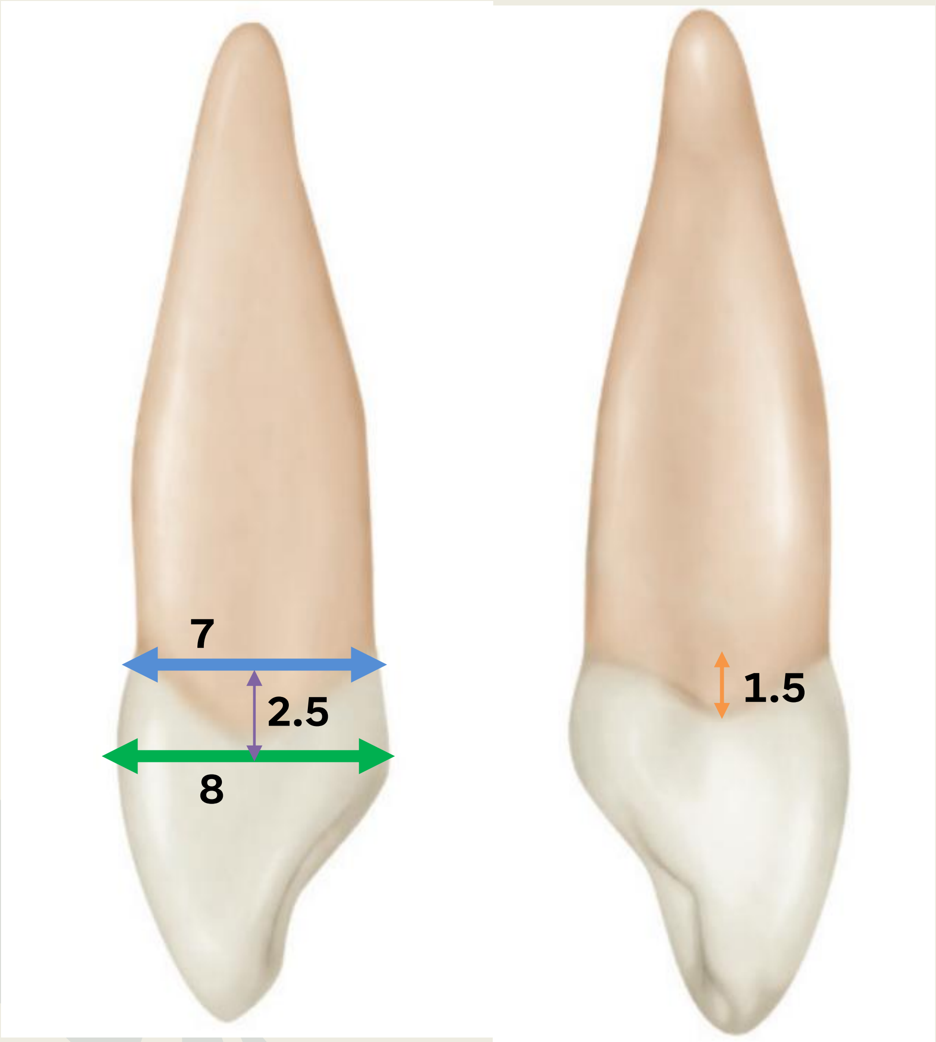
FIRST EVIDENCE OF CALCIFICATION	4–5 MO
CROWN COMPLETION	6-7 years
ERUPTION	11-12 years
ROOT COMPLETION	13-15 years

DIMENSIONS OF MAXILLARY CANINE

Cervico-incisal length of crown	10 mm
Length of root	17 mm
Mesiodistal Diameter of Crown	7.5 mm
Mesiodistal Diameter of Crown at Cervix	5.5 mm

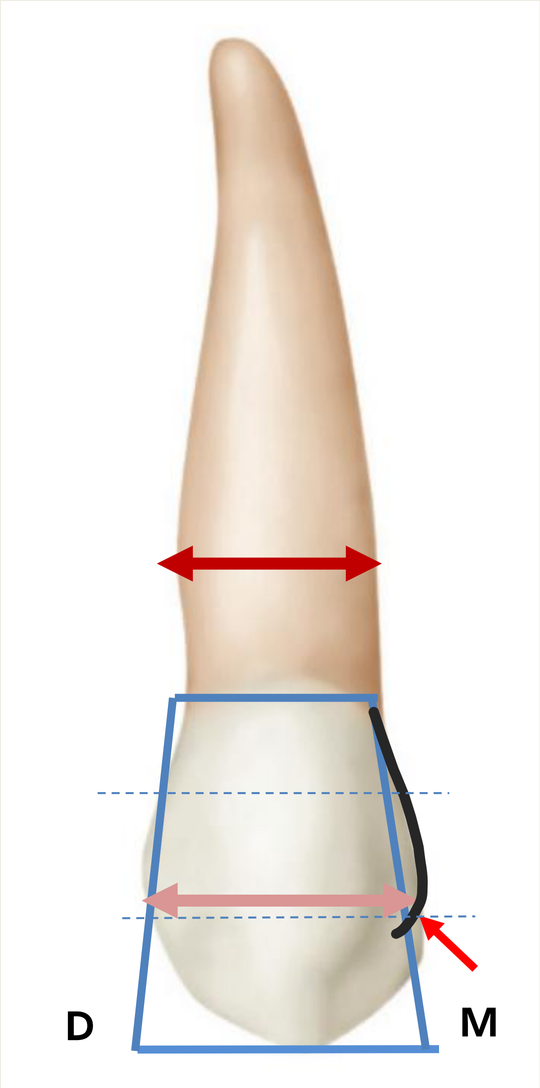
DIMENSIONS OF MAXILLARY CANINE

Labiolingual Diameter of crown	8 mm
Labiolingual Diameter of crown at cervix	7 mm
Curvature of Cervical line (M)	2.5 mm
Curvature of Cervical line (D)	1.5 mm



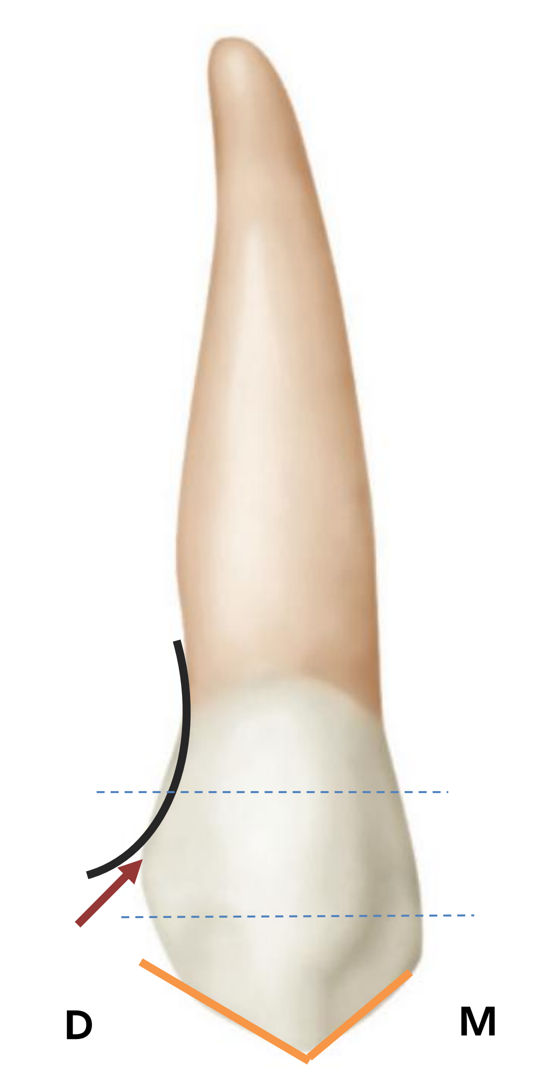
LABIAL ASPECT

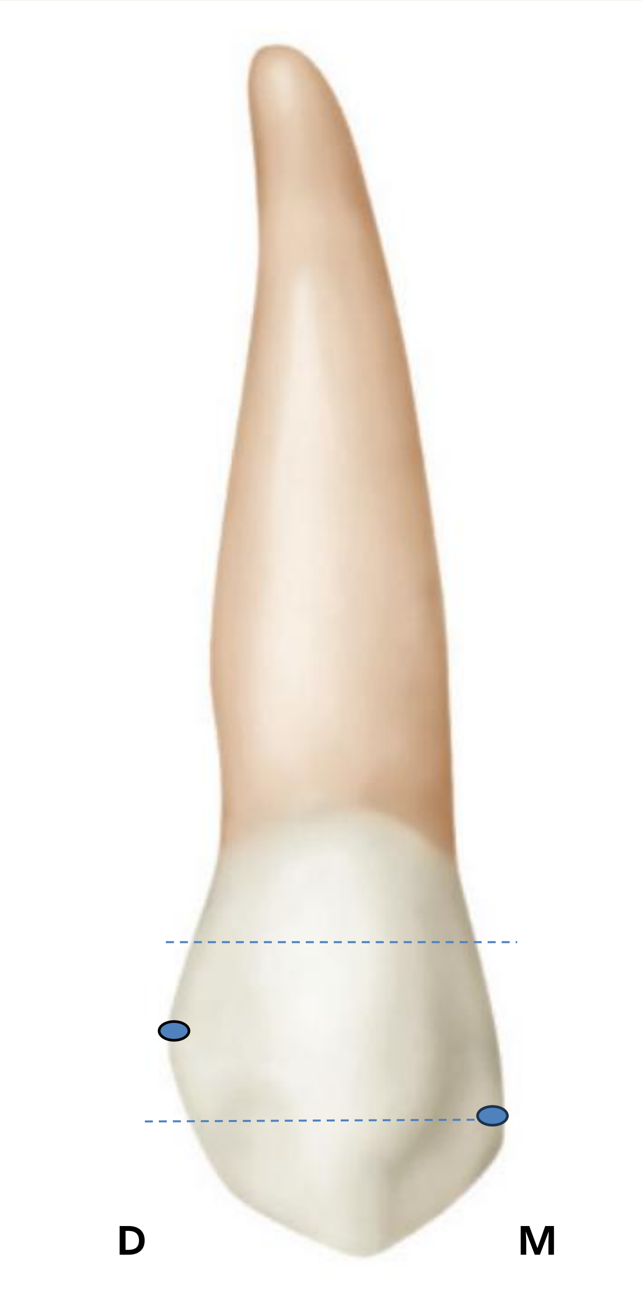
- From facial aspect, **trapezoidal** crown form.
- The crown and root are narrower mesiodistally than those of the maxillary central incisor.
- Mesially, the outline of the crown is convex from the cervix to the center of the mesial contact area.
- Contact area mesially is approximately at the junction of middle and incisal thirds.



LABIAL ASPECT

- Distally, the outline of the crown is usually concave between the cervical line and the distal contact area.
- The distal contact area is at the center of middle third.
- A concave outline in the distolabial cervical third.
- The cusp has a mesial and a distal slope. The mesial cuspal ridge is shorter than the distal cuspal ridge.



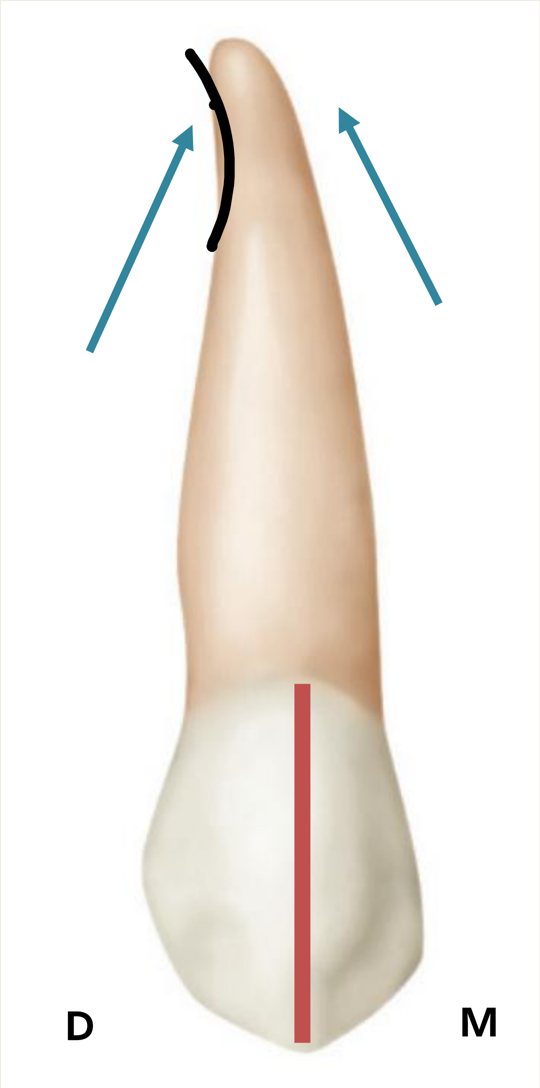


CONTACT AREAS

Distal- Center of the middle third
Mesial- Junction of the middle and incisal third

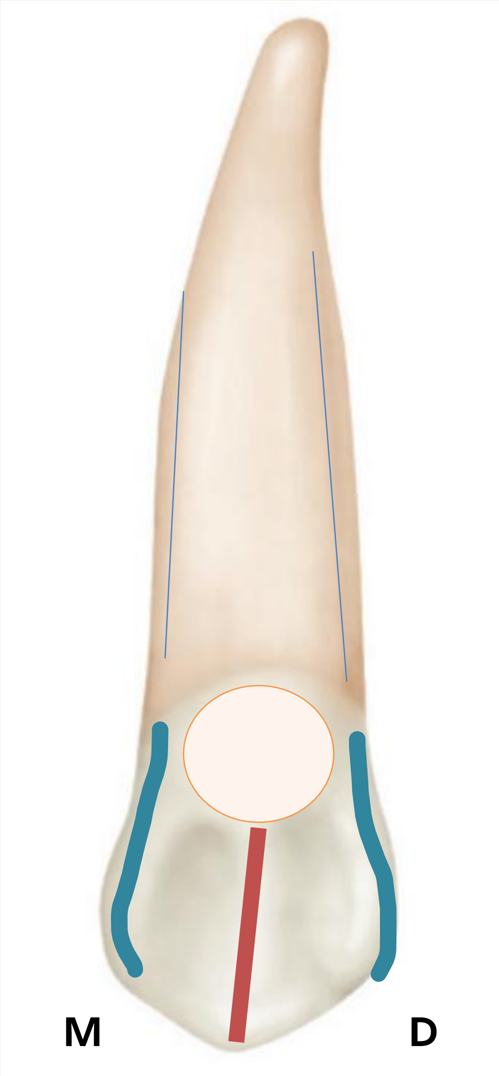
LABIAL ASPECT

- The cusp tip is in line with the center of the root.
- The middle labial lobe shows much greater development. This produces a ridge: **the labial ridge** (extends from the cusp tip to cervical ridge).
- The root appears slender from the labial aspect; it is conical with a bluntly pointed apex.
- The root has a sharp curve in the vicinity of the apical third (mostly distally).



LINGUAL ASPECT

- The crown and root are narrower than labially.
- The cingulum is large and, in some instances, is pointed like a small cusp.
- A well-developed lingual ridge is seen that is confluent with the cusp tip; this extends to a point near the cingulum.
- Mesial and distal marginal ridges mark the boundaries.



LINGUAL ASPECT

C: cingulum

MMR: mesial marginal ridge

DMT: distal marginal ridge

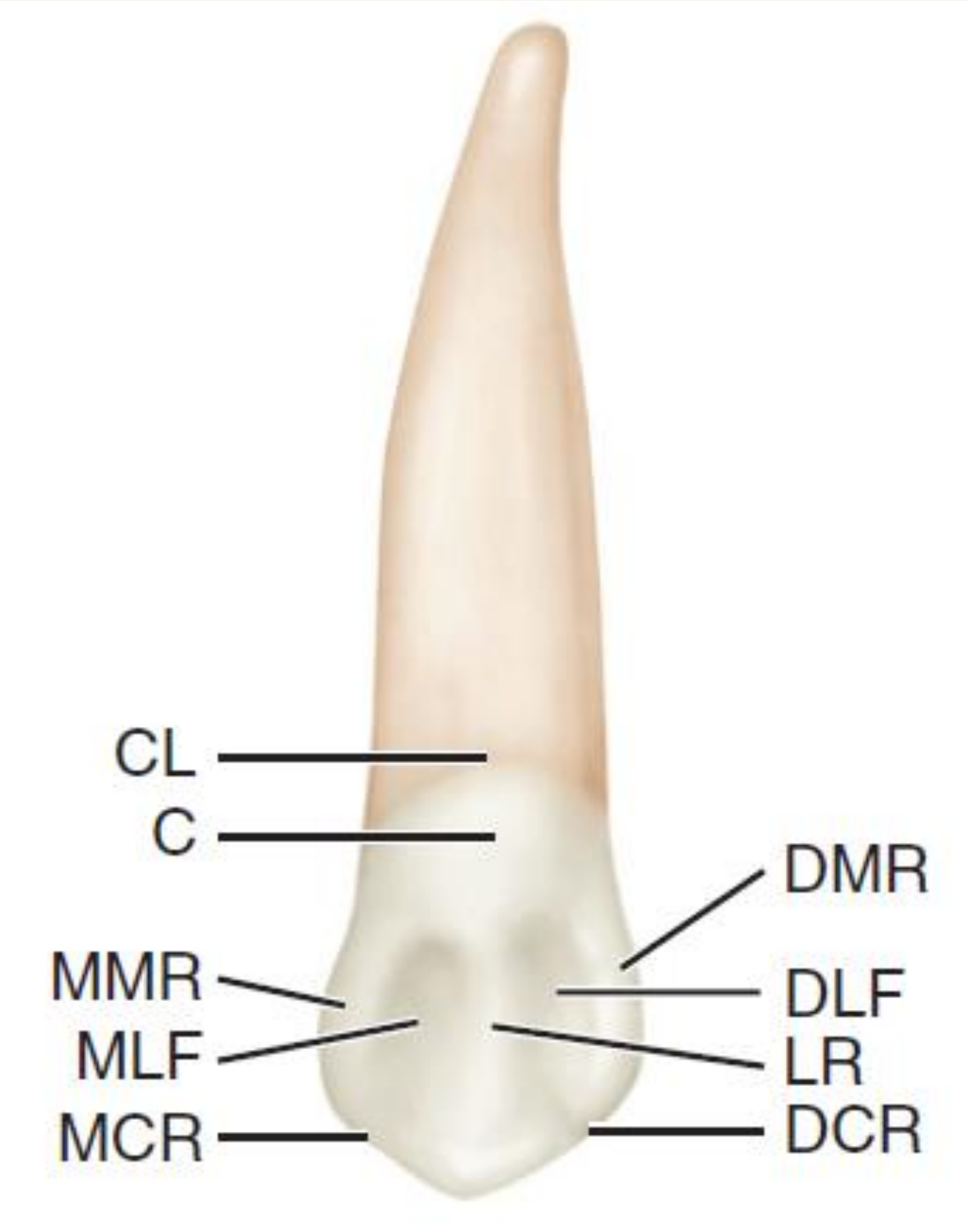
MCR: mesial cuspal ridge

DCR: distal cuspal ridge

LR: lingual ridge

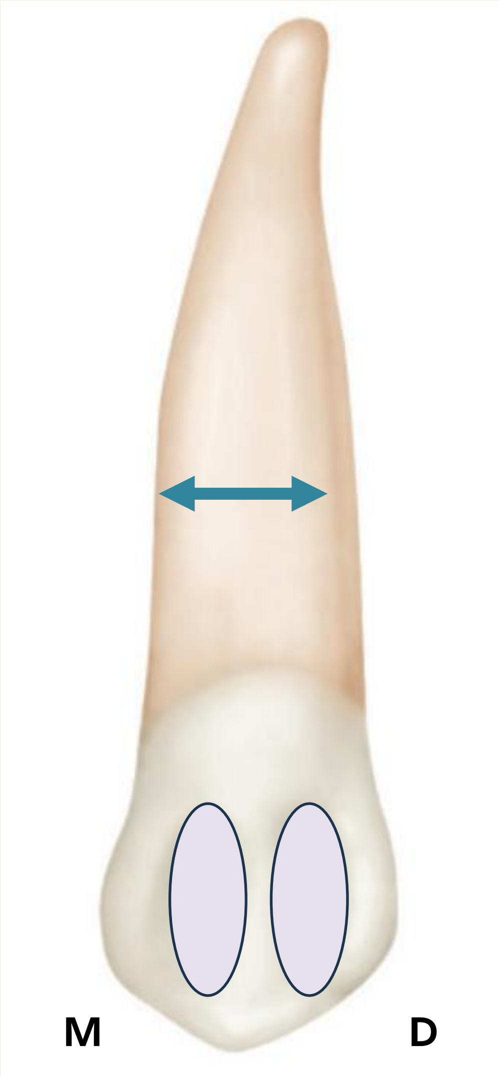
MLF: mesial lingual fossa

DLF: distal lingual fossa



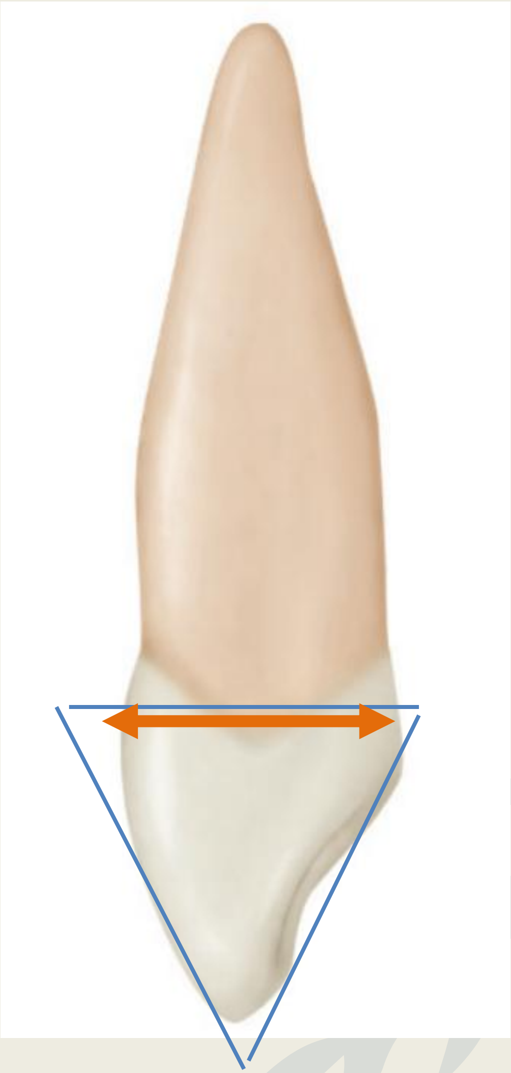
LINGUAL ASPECT

- Shallow concavities are evident between lingual ridge and the marginal ridges—the mesial and distal lingual fossae.
- The root is conical- the longest of all roots and tapers lingually.
- The lingual portion of the root is narrower than the labial portion.



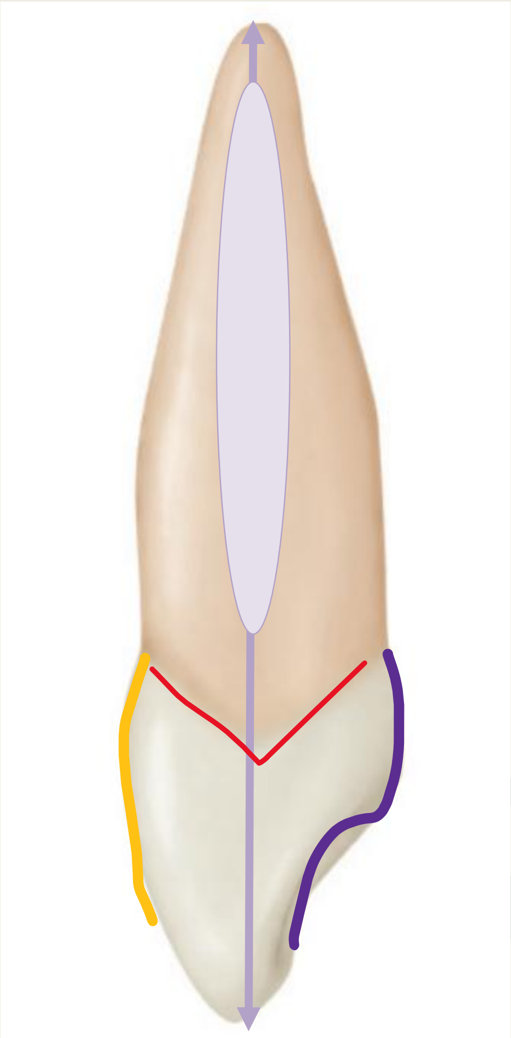
MESIAL ASPECT

- The crown form is considered triangular/wedge-shaped, like all the anterior teeth.
- This tooth has the greatest labiolingual measurement of any anterior tooth.
- The crown is widest at the crests of the cervical ridges.



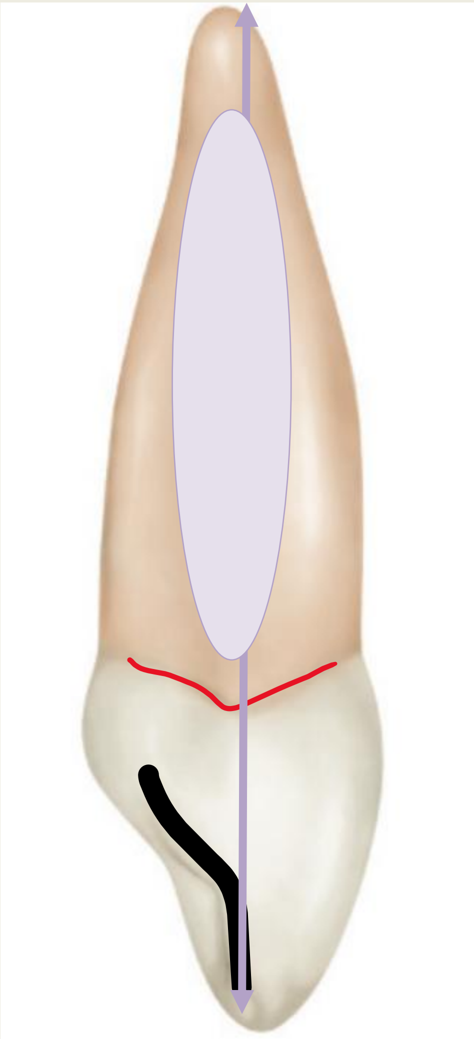
MESIAL ASPECT

- The cusp tip, height of curvature of the cervical line and the root apex are all aligned.
- Facial outline is convex in the cervical third, then flat middle third, and convex in the incisal third.
- The root is wide with a developmental depression present throughout most of the root- help prevent rotation and displacement.



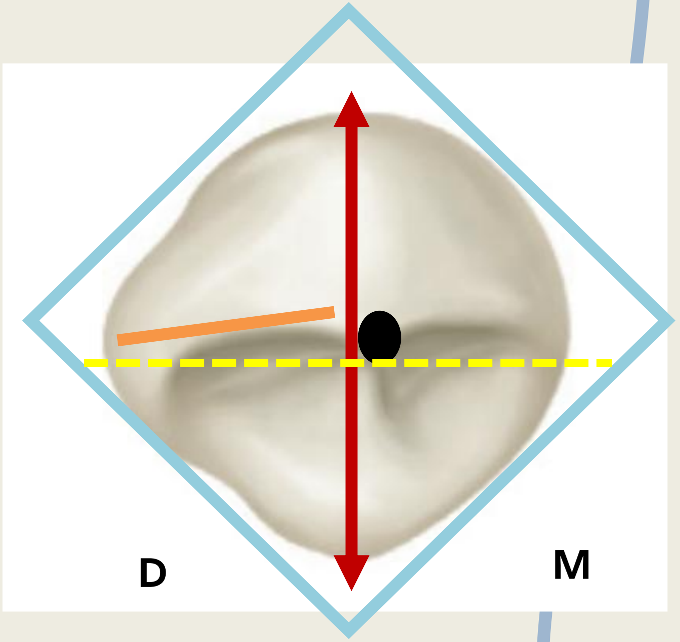
DISTAL ASPECT

- The cervical line exhibits less curvature toward the cusp ridge.
- The distal marginal ridge is heavier and more irregular in outline.
- The cusp tip is aligned with the root tip, and the height of curvature of the cervical line.
- The root surface developmental depression on the distal is deeper than mesial.



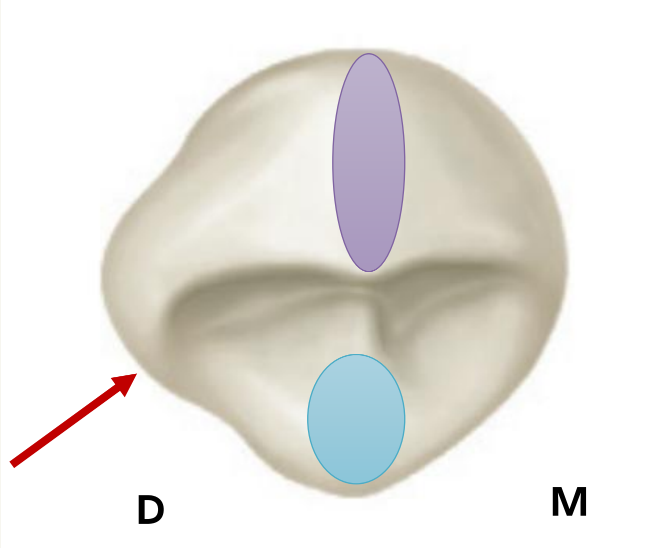
INCISAL ASPECT

- The crown outline is diamond shaped.
- Labiolingual dimension is greater than the mesiodistal.
- The cusp tip is labial to the center of the crown labiolingually and mesial to the center mesiodistally.
- Distal slope is longer than mesial slope.

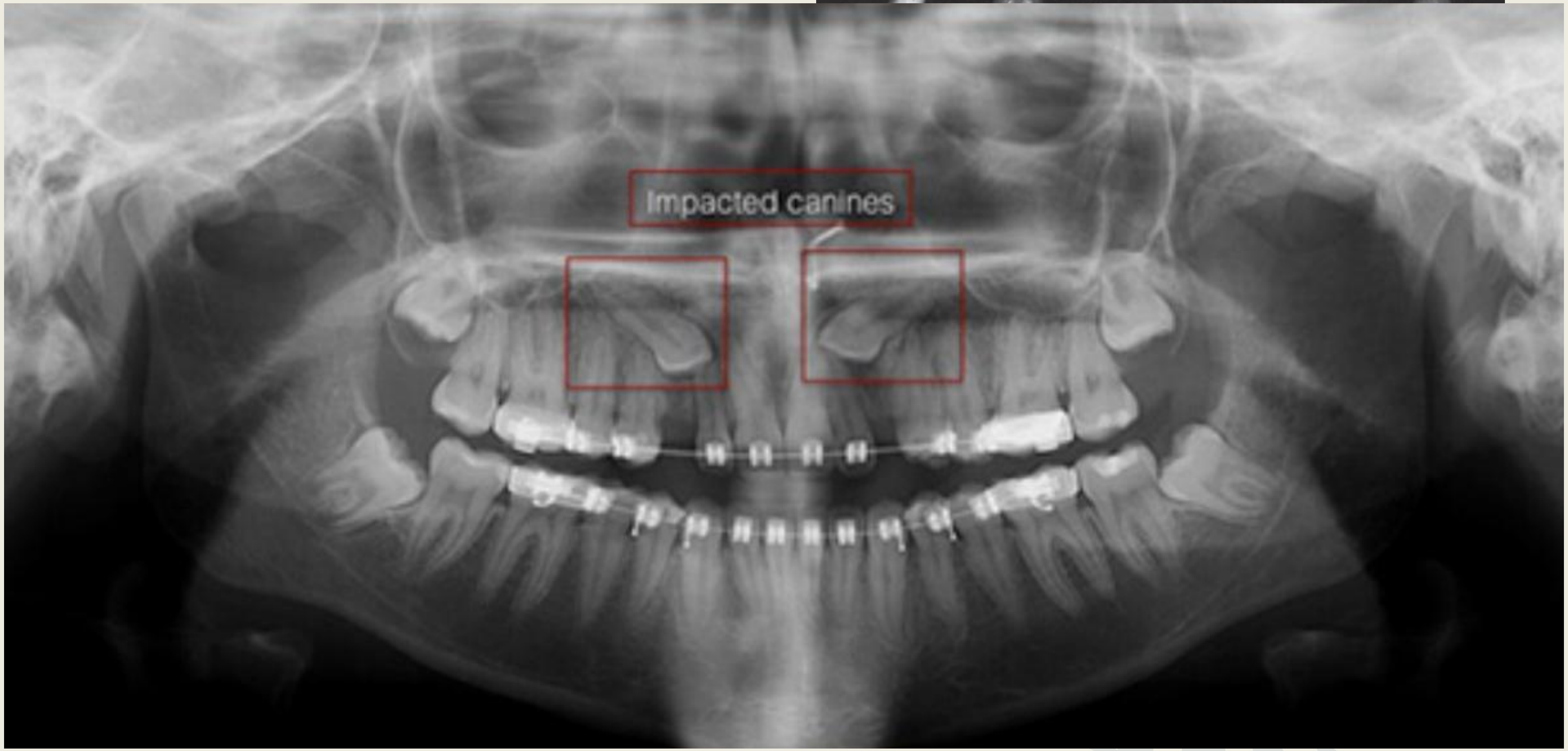


INCISAL ASPECT

- Distal part appears "stretched" to contact the maxillary first premolar.
- The ridge of the middle labial lobe is very noticeable labially from this aspect. Greatest convexity at the cervical third.
- The cingulum makes up the entire cervical third of crown.



VARIATIONS AND CLINICAL IMPLICATIONS



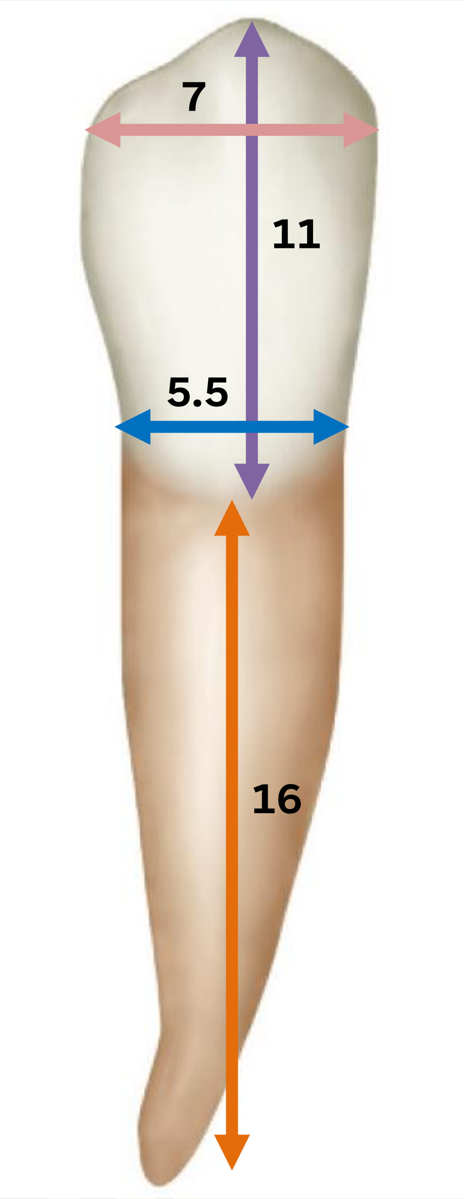
MORPHOLOGY OF MANDIBULAR CANINES

CHRONOLOGY OF MANDIBULAR CANINE

FIRST EVIDENCE OF CALCIFICATION	4–5 MO
CROWN COMPLETION	6-7 years
ERUPTION	9-10 years
ROOT COMPLETION	12-14 years

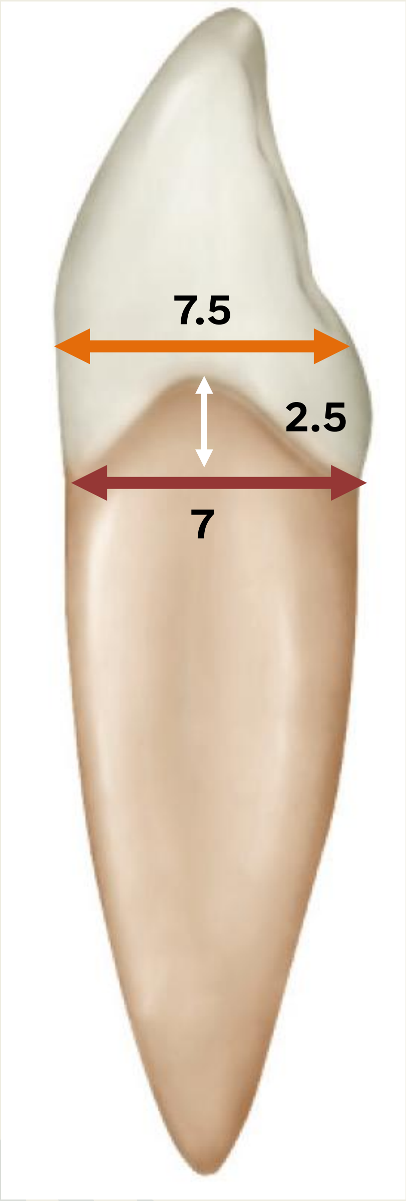
DIMENSIONS OF MANDIBULAR CANINE

DIMENSIONS	MAX CANINE	MAND CANINE
Cervico-incisal length of crown	10 mm	11 mm
Length of root	17 mm	16 mm
Mesiodistal Diameter of Crown	7.5 mm	7 mm
Mesiodistal Diameter of Crown at Cervix	5.5 mm	5.5 mm



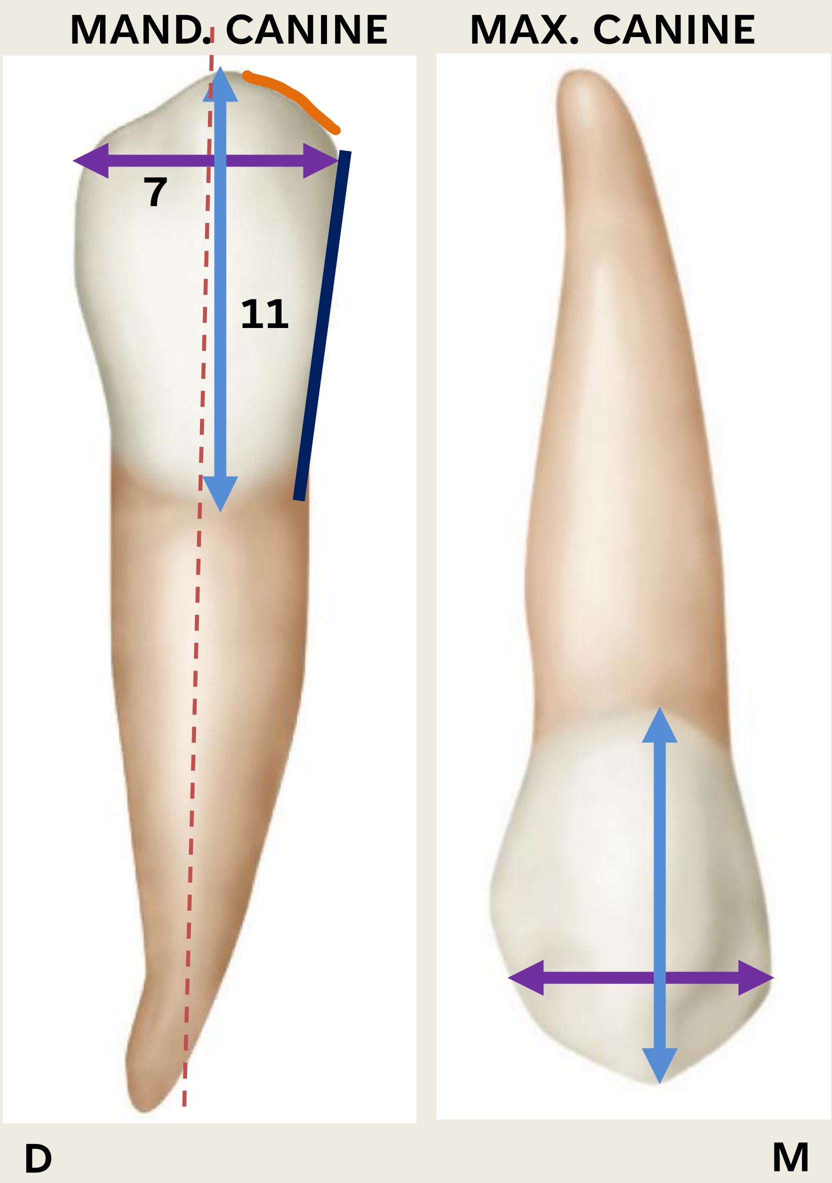
DIMENSIONS OF MANDIBULAR CANINE

DIMENSIONS	MAX CANINE	MAND CANINE
Labiolingual Diameter of crown	8 mm	7.5 mm
Labiolingual Diameter of crown at cervix	7 mm	7 mm
Curvature of Cervical line (M)	2.5 mm	2.5 mm
Curvature of Cervical line (D)	1.5 mm	1 mm



LABIAL ASPECT

- Crown is narrower mesiodistally than maxillary and longer by 1mm.
- Crown is broader mesiodistally than either of the mandibular incisors.
- The mesial outline of the crown is straight, mesial cuspal ridge shorter.
- Cusp angle is in line with the center of the root. (same as max.)



LABIAL ASPECT

- Distal outline of the crown is slightly rounded.
- A labial ridge is present but not as well developed.
- Cervical line labially has a semicircular curvature apically.
- The root is shorter than that of maxillary canine, and its apical end more pointed.

D

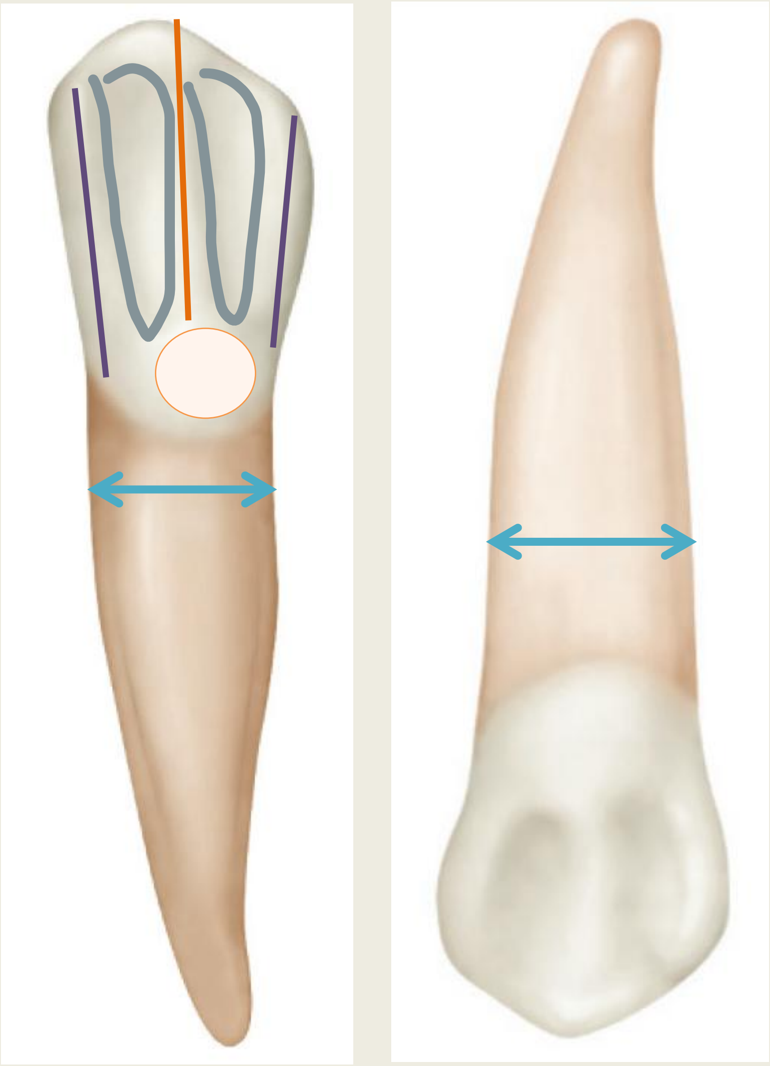
M

CONTACT AREAS

Distal- middle third (more incisally)
Mesial- incisal third

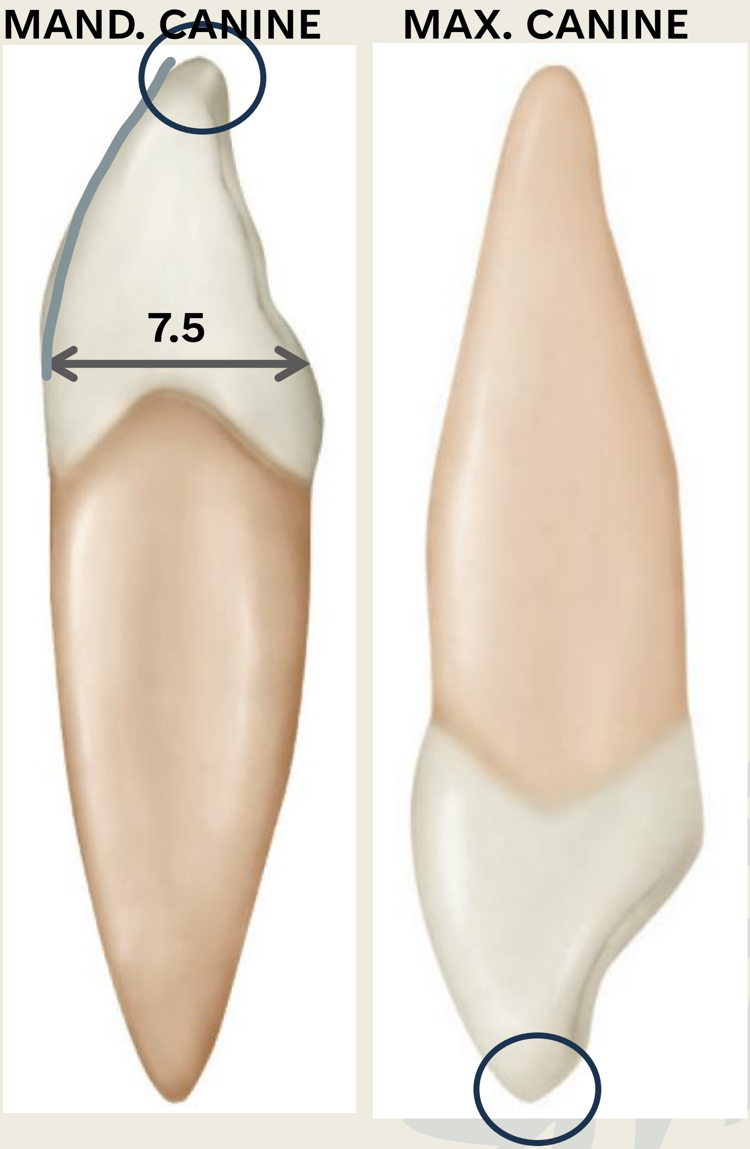
LINGUAL ASPECT

- Lingual surface **is flatter, smooth and regular** than maxillary but resembles mandibular incisors.
- The lingual portion of the root is narrower than the maxillary.
- Cingulum: poorly developed.
- Marginal ridges: less prominent.
- Lingual ridge: less prominent
- Mesial and distal linguo fossae: less deep



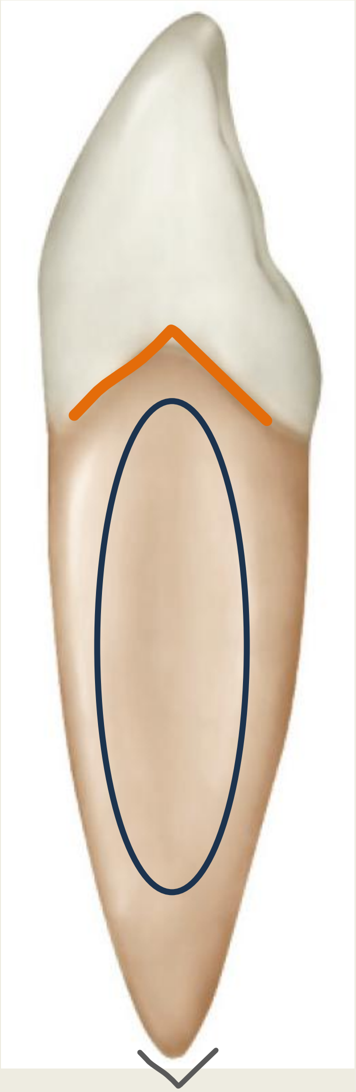
MESIAL ASPECT

- The crown form is triangular/wedge-shaped, like all the anterior teeth.
- Labiolingual measurement is less than maxillary.
- It has less curvature labially on the crown.
- Cingulum is not as pronounced, and the incisal portion is thinner labiolingually, the cusp appears more pointed.



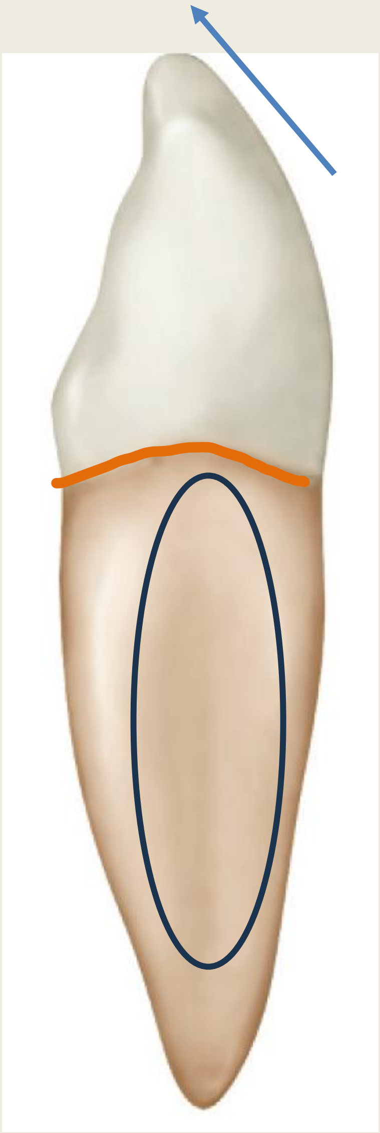
MESIAL ASPECT

- The cervical line curves more toward the incisal portion.
- The roots is similar with exception of a more pointed root tip on the mandibular canine.
- The developmental depression mesially is prounounced and deep.



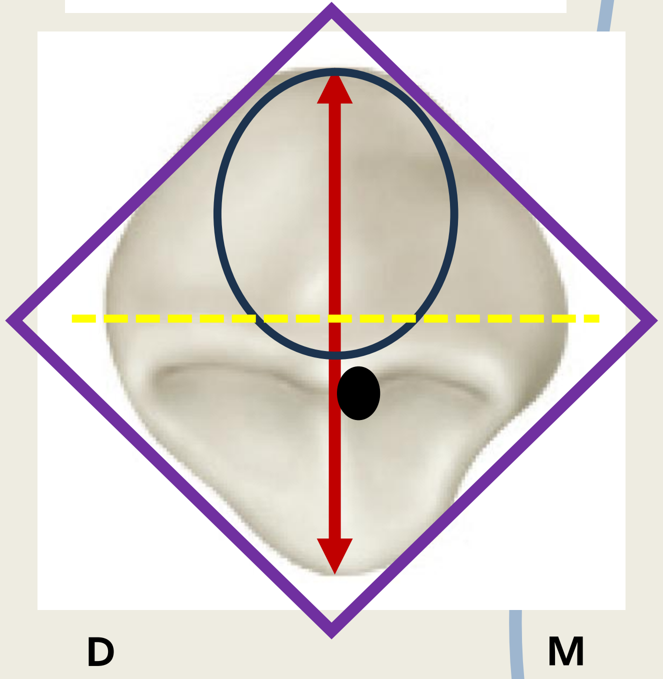
DISTAL ASPECT

- There is very little difference in comparison to the mesial view.
- The height of curvature of cervical line toward incisal is flatter
- Developmental root depression is present.
- Cusp tip is lingually inclined with respect to the root axis.



INCISAL ASPECT

- The crown outline diamond shaped.
- Labiolingual measurement is greater than mesiodistal.
- The crown is bulkier on the labial side.
- Cusp tip and cuspal ridges are lingual to the bisector line. (the only major difference)



MORPHOLOGICAL VARIATIONS

Root bifurcation (labial and lingual roots) is a rare variation of mandibular canine.

DIFFERENCES BETWEEN MAXILLARY AND MANDIBULAR CANINES

	MAXILLARY CANINE	MANDIBULAR CANINE
DIMENSIONS	Crown is wider mesiodistally	Crown is narrower mesiodistally but longer cervicoincisally
LABIAL VIEW	Mesial outline is convex Distal outline is concave The mesial cuspal ridge is shorter than the distal cuspal ridge. Pronounced labial ridge Distal contact area: center of middle third Mesial- Junction of middle and incisal third	Mesial outline is straight Distal outline is rounded The mesial cuspal ridge is shorter than the distal cuspal ridge Less pronounced labial ridge Distal contact area: middle third (more incisally) Mesial contact area: incisal third (mesioincisal line angle)
LINGUAL VIEW	Prominent and pronounced lingual anatomical landmarks	Smoother and less pronounced lingual landmarks

DIFFERENCES BETWEEN MAXILLARY AND MANDIBULAR CANINES

	MAXILLARY CANINE	MANDIBULAR CANINE
PROXIMAL VIEWS	Wide labiolingually Height of contours pronounced Blunt cusp tip, inclined labially	Labiolingual measure is lesser than maxillary. Height of contours less pronounced Sharp cusp tip, inclined more lingually
INCISAL VIEW	Greater labiolingual than MD Cusp is mesially deviated	Much greater labiolingual Cusp is mesially deviated
ROOT	17mm Blunt root tip Developmental depression is pronounced	1mm shorter Pointed root tip Developmental depression mesially is pronounced and deep

Q- In comparison with the mandibular canine, the maxillary canine:

A. Is wider mesiodistally
B. Has a shorter cingulum
C. Has apex deviating distally
D. Erupts before mandibular canine
E. Has sharper cuspal slopes

Thank you!